

DIFFERENTIAL GEOMETRY

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ABSTRACT. Differential Geometry is regarded as a branch of mathematics that uses calculus and linear algebra to study the shapes, curves, and surfaces of smooth curved spaces. Unlike elementary geometry, which studies flat surfaces and rigid shapes, differential geometry focuses on local and global characteristics of curved spaces where geometric properties vary continuously from point to point. This framework serves as the mathematical language for modern theoretical physics.

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1. A LIGHT TREATMENT ON THE GLOBAL PROPERTIES OF CURVES

1.1. Global Properties of Plane Curves

We explore briefly 3 main topics (of increasing difficulty):

1. Isoperimetric inequality
2. Four-Vertex Theorem
3. Cauchy-Crofton Formula

Some helpful definitions to note before beginning:

Definition 1.1. A differentiable function on a closed interval $[a, b]$ is the restriction of a differentiable function defined on an open interval containing $[a, b]$.

Definition 1.2. A closed plane curve is a regular parametrized curve $\alpha: [a, b] \rightarrow \mathbb{R}^2$ such that α and all its derivatives agree at a and b

Note that *simple* curves are ones without intersections.



Figure 1: Vector field paths along curves.

1.2. The Isoperimetric Inequality

This seeks to address the following problem: *of all simple closed curves in the plane with a given length l , which one bounds the largest area?*

Let A be an area bounded by a positively oriented simple closed curve $\alpha(t) = (x(t), y(t))$, $t \in [a, b]$:

$$A = - \int_a^b y(t)x'(t) dt = \int_a^b x(t)y'(t) dt = \frac{1}{2} \int_a^b (xy' - yx') dt \quad (1)$$

The second formula is obtained by the first:

$$\begin{aligned} \int_a^b xy' dt &= \int_a^b (xy)' dt - \int_a^b x'y dt = [xy(b) - xy(a)] - \int_a^b x'y dt \\ &= \int_a^b x'y dt, \end{aligned}$$

To present a simple example, let:

$$y = f_1(x) \quad \text{and} \quad y = f_2(x), \quad x \in [x_0, x_1], \quad f_1 > f_2. \quad (2)$$

To illustrate:

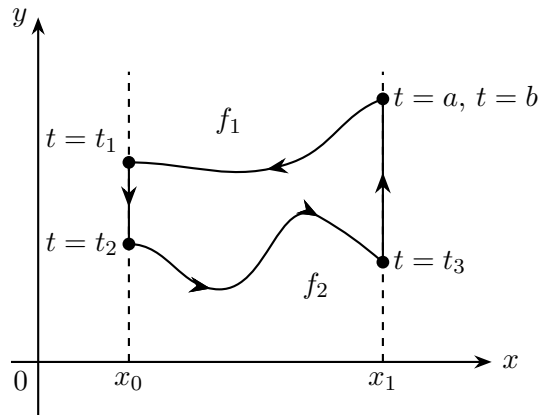


Figure 2: Parametrization flow over a bounded region.

Recall from a calculus II class that in order to solve for the area between the curves, we find:

$$A = \int_{x_0}^{x_1} f_1(x) dx - \int_{x_0}^{x_1} f_2(x) dx. \quad (3)$$

Since the curve is positively oriented,

$$A = - \int_a^{t_1} y(t)x'(t) dt - \int_{t_2}^{t_3} y(t)x'(t) dt = - \int_a^b y(t)x'(t) dt. \quad (4)$$

Rather than this approach, we can also find the results of eq. (1) using Green's theorem in the plane.

Theorem 1.3 (The Isoperimetric Inequality). *Let C be a simple closed plane curve with length l , and let A be the area of the region bounded by C . Then*

$$l^2 - 4\pi A \geq 0, \quad (5)$$

and equality holds if and only if C is a circle.

REFERENCES

- [1] Manfredo P. do Carmo, *Differential Geometry of Curves and Surfaces*, Revised & Updated 2nd ed., Dover Publications, 2016.
- [2] Barrett O'Neill, *Elementary Differential Geometry*, Revised 2nd ed., Academic Press, 2006.